

COMPLETE LINUX INSTALLATION & DEVELOPER SETUP GUIDE

Why I Chose Linux Over Windows — Installation, Configuration & Development Workflow

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Personal Learning Document | Beginner-Friendly Guide

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1. Introduction

Linux is an open-source operating system widely used by developers, cybersecurity professionals, researchers, system administrators, and software engineers. It is known for its flexibility, performance, customization, stability, and developer-friendly environment.

As a student developer, I wanted an operating system that could provide:

- Better development workflows and improved performance
- Deep customization and terminal access
- Open-source tools and a distraction-free learning environment

This document contains a complete beginner-friendly Linux installation and setup guide based on my personal experience exploring Linux and developer workflows.

2. Why I Chose Linux Over Windows

Windows is suitable for general users, gaming, and compatibility with commercial software. However, while learning software development, backend technologies, open-source tools, Docker, Linux commands, and system-level workflows, Linux offered more control and flexibility.

Key reasons for choosing Linux:

- Developer-friendly, highly customizable and lightweight
- Efficient for programming with better terminal-based workflows
- Strongly connected to open-source development

Skills gained through Linux exploration:

- Linux commands, shell environments, and Git workflows
 - Development setup, system configuration, and real-world developer environments
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3. Important Preparations Before Installation

Before installing Linux, it is strongly recommended to complete the following preparations.

Backup Important Data

Always backup all critical files before proceeding. Recommended backup methods:

- External hard drive or USB storage
 - Cloud services: Google Drive, OneDrive
 - GitHub repositories for code and projects
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4. Disable BitLocker (Very Important)

 **Note:** If BitLocker encryption is enabled, Linux installation may fail and Windows partitions may become inaccessible.

Step 1: Open Control Panel

Step 2: Navigate to: System and Security → BitLocker Drive Encryption

Step 3: Check BitLocker status. If enabled, select Turn Off BitLocker.

Step 4: Wait for the full decryption process to complete before continuing.

5. Disable Fast Startup

Windows Fast Startup may cause dual boot issues, filesystem corruption, and partition mounting problems.

Step 1: Open Control Panel

Step 2: Go to: Power Options

Step 3: Select: Choose what the power buttons do

Step 4: Disable: Turn on fast startup, then save changes.

6. Download Linux ISO

Recommended beginner-friendly Linux distributions:

Distribution	Official Website
Fedora	https://fedoraproject.org
Ubuntu	https://ubuntu.com
Nobara	https://nobaraproject.org

Download the latest ISO file from the official website of your chosen distribution.

7. Create Bootable USB

A bootable USB drive is required for Linux installation.

Recommended Tools

Tool	Website
Rufus	https://rufus.ie
Balena Etcher	https://etcher.balena.io

Recommended Rufus Settings

Setting	Recommended Value	Notes
Partition Scheme	GPT	For modern UEFI systems
Target System	UEFI	Not Legacy
File System	FAT32	Standard for bootable USB

8. BIOS & Boot Menu Configuration

Enter BIOS

Use the following keys during startup to enter the BIOS:

Laptop Brand	BIOS Key
HP	ESC / F10
Dell	F12
Lenovo	F12
ASUS	ESC
Acer	F12

BIOS Settings to Verify

- Enable USB Boot — ensure USB Boot is enabled in the boot order.
- Use UEFI Mode — UEFI mode is recommended instead of Legacy Mode.
- Disable Secure Boot (if necessary) — Secure Boot may interfere with Linux installation or NVIDIA drivers.

9. Dual Boot Preparation

Shrink Windows Partition

Step 1: Right-click the Start Button

Step 2: Open: Disk Management

Step 3: Right-click C: Drive and select Shrink Volume

Recommended Storage Allocation:

Usage Type	Recommended Space
Basic Linux Usage	50 GB
Development Setup	100 GB
Gaming + Development	150 GB

After shrinking the partition, unallocated space will be available for Linux installation.

10. Linux Installation Process

Boot Into USB

- Restart the system and open the Boot Menu
- Select: USB Flash Drive
- From the Linux boot screen, choose: Try or Install Linux

Installation Type

For Dual Boot — Select: Install alongside Windows

For Linux-Only Installation — Select: Erase Disk and Install Linux

Warning: The 'Erase Disk' option permanently removes ALL existing data. Ensure backups are complete before proceeding.

User Setup

Configure your username, password, and computer name, then continue installation until completion.

11. First Boot & Initial Setup

After installation: remove the USB drive, restart the system, and boot into Linux.

System Update Commands

Fedora / Nobara:

```
sudo dnf update -y
```

Ubuntu:

```
sudo apt update
sudo apt upgrade -y
```

12. Essential Linux Commands

Command	Purpose
pwd	Show current working directory
ls	List files and directories
cd	Change directory
mkdir	Create a new directory
rm	Remove files or directories
clear	Clear the terminal screen
sudo	Run command with administrator privileges

13. Developer Environment Setup

The following essential tools form the foundation of a modern developer environment on Linux:

Tool	Purpose
Git	Version Control — track and manage code changes
VS Code	Code Editor — feature-rich, extensible editor
Docker	Containers — package and run applications in isolated environments
Node.js	JavaScript Runtime — server-side JavaScript execution
Python	Programming — scripting, data science, automation
Postman	API Testing — test and document REST APIs

Install VS Code from the official website: <https://code.visualstudio.com>

14. Recommended Applications

Application	Purpose
Flameshot	Screenshots — powerful annotation tool
OBS Studio	Screen Recording — professional streaming and recording
VLC	Media Playback — supports virtually all media formats
GParted	Partition Management — graphical disk partition editor

Application	Purpose
Neofetch	System Information — displays system stats in terminal
Flatseal	Flatpak Permission Control — manage app permissions

15. Recommended VS Code Extensions

Extension	Purpose
Prettier	Automatic code formatting
ESLint	JavaScript/TypeScript linting
GitLens	Advanced Git visualization and history
Material Icon Theme	Better file icons in the explorer
Error Lens	Inline error and warning highlighting
Tailwind CSS IntelliSense	Tailwind CSS class autocompletion
Python	Python language support and debugging
Java Extension Pack	Complete Java development support
Docker	Docker integration and container management
Thunder Client	Lightweight REST API testing client

16. Recommended GNOME Extensions

Install from the official website: <https://extensions.gnome.org>

Extension	Benefit
Blur My Shell	Adds blur effects to the shell and dock
Dash to Dock	Converts the dash into a taskbar-style dock
User Themes	Enables custom GNOME shell themes
Clipboard Indicator	Clipboard history and manager
GSConnect	Connect Android phone to Linux desktop
AppIndicator	System tray support for legacy apps

These extensions collectively improve productivity, appearance, workflow efficiency, and desktop customization.

17. Common Problems & Fixes

WiFi Not Working

- Update the system: `sudo dnf update`
- Install required drivers if the issue persists after updating.

```
sudo dnf update
```

Black Screen After Boot

Possible causes:

- NVIDIA drivers not installed or conflicting
- Secure Boot interfering with drivers
- Corrupted ISO file during installation

Windows Missing in Dual Boot

Run the following command to regenerate the GRUB boot menu:

```
sudo update-grub
```

USB Not Detected

- Recreate the bootable USB using Rufus or Balena Etcher
- Try a different USB port (prefer USB 2.0 ports for compatibility)
- Disable Secure Boot in BIOS settings

18. My Experience Using Linux

Exploring Linux helped me build a deeper understanding of how operating systems and development environments work. Working through the installation, configuration, and daily usage provided hands-on experience that theoretical study alone cannot replicate.

Key areas of growth through Linux:

- Terminal workflows and command-line confidence
- System configuration and package management
- Open-source tools and developer ecosystem
- Git workflows and version control best practices
- Debugging skills and problem-solving mindset

Compared to Windows, Linux felt more efficient, more customizable, and more aligned with professional developer environments.

19. Final Thoughts

Linux is not just an operating system for developers — it is also a learning environment that encourages exploration, customization, and technical growth.

Although the installation process may initially seem challenging for beginners, learning Linux provides invaluable experience in:

- Development workflows and system management
- Terminal usage and command-line proficiency
- Open-source technologies and community tools
- Software engineering environments and real-world workflows

This guide is based on personal learning experience while exploring Linux, development tools, and modern developer workflows as a student developer.

— **Sravanya**